CS 374: Section A - Homework 4, Problem 2

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Throughout, $\Sigma = \{0, 1\}$, and $\#(a, x)$ denotes the number of occurrences of the symbol a in the string x.

Part (a)

All strings in which every prefix has at least as many 1s as 0s, that is,

$$\{w \in \Sigma^* \mid \#(1, x) \geq \#(0, x) \text{ for every prefix } x \text{ of } w\}.$$

$S \to \varepsilon \mid 1S \mid 1B0S$	every prefix has at least as many 1s as 0s
$B \to \varepsilon \mid 1B0B$	balanced, every prefix has at least as many 1s as 0s

For a string u write $d(u) = \#(1, u) - \#(0, u)$. The two non-terminals generate the following languages.

- B generates every string u with $d(u) = 0$ such that $d(x) \geq 0$ for every prefix x of u. These are the strings with equally many 0s and 1s in which no prefix has more 0s than 1s.
- S generates every string w such that $d(x) \geq 0$ for every prefix x of w, which is exactly the language asked for.

Why this is correct. Reading a 1 raises d by one and reading a 0 lowers it by one, so d of a prefix is the running count that the language constrains. Each description is checked in both directions.

Every string B generates is balanced with non-negative prefix counts. This is clear for ε. If $u, v \in B$, then the prefixes of $1u0v$ are the strings $1u'$, with $d = 1 + d(u') \geq 1$, and the strings $1u0v'$, with $d = 1 + 0 - 1 + d(v') \geq 0$. The whole string has $d(1u0v) = 0$.

Every such string is generated by B. Let u be non-empty with $d(u) = 0$ and $d(x) \geq 0$ for every prefix x. Its first symbol is 1, since the prefix 0 would have $d = -1$. Let i be the smallest index with $d(u[1..i]) = 0$; this index exists because $d(u) = 0$, and $i \geq 2$. Then $u[i] = 0$, because $d(u[1..i - 1]) \geq 1$ by the choice of i. So

$$u = 1y0z, \quad \text{where } y = u[2..i - 1] \text{ and } z = u[i + 1..].$$

Here $d(y) = 0$, and every prefix y' of y has $d(y') = d(1y') - 1 \geq 0$, again by the choice of i, so $y \in B$. And z is the rest of u after a prefix with $d = 0$, so $d(z) = 0$ and every prefix of z has $d \geq 0$, so $z \in B$. This is the rule $B \to 1B0B$, and induction on length finishes the argument.

Every string S generates has non-negative prefix counts. This is clear for ε. If $w \in S$, then every prefix of $1w$ has $d = 1 + d(w') \geq 1$. If $u \in B$ and $w \in S$, then the prefixes of $1u0w$ have $d \geq 1$, then $d = 0$, then $d = d(w') \geq 0$.

Every such string is generated by S. Let w be non-empty with $d(x) \geq 0$ for every prefix x, and apply the hint.

- If no non-empty prefix x of w has $d(x) = 0$, then every non-empty prefix has $d \geq 1$. In particular $w = 1v$, and every prefix v' of v has $d(v') = d(1v') - 1 \geq 0$, so $v \in S$ and w comes from $S \to 1S$.
- Otherwise let x be the shortest non-empty prefix with $d(x) = 0$. Exactly as in the argument for B, we have $x = 1y0$ with $y \in B$, and the remaining suffix w' has $d(w'') = d(xw'') \geq 0$ for each of its prefixes w'', so $w' \in S$ and w comes from $S \to 1B0S$.

Induction on length finishes the argument.

Sources and collaborators: None.

LLM use: I used Claude (Anthropic) to draft these grammars and explain their structure to me. I checked every grammar by hand against the examples in the handout and against several strings inside and outside each language, and typeset the submission myself.

Part (b)

Even-length strings whose second half contains an even number of 1s, that is,

$$\{xy \in \Sigma^* \mid |x| = |y| \text{ and } \#(1, y) \text{ is even}\}.$$

$X \to 0 \mid 1$	the two one-symbol strings
$E \to \varepsilon \mid XE0 \mid XO1$	even-length, second half has an even number of 1s
$O \to XO0 \mid XE1$	even-length, second half has an odd number of 1s

The start symbol is E. The three non-terminals generate the following languages.

- X generates the two strings 0 and 1.
- E generates every string xy with $|x| = |y|$ and $\#(1, y)$ even.
- O generates every string xy with $|x| = |y|$ and $\#(1, y)$ odd.

Why this is correct. Every rule other than $E \to \varepsilon$ adds one symbol at the far left and one symbol at the far right, so a derivation of a string of length $2n$ applies exactly n of them. The n symbols added on the left, in the order they were added, spell out the first half x, and the n symbols added on the right spell out the second half y from its last symbol backwards. The halves therefore always have equal length, and the only thing the rules need to track is the parity of $\#(1, y)$.

- A symbol 0 placed on the right keeps that parity, which is why $E \to XE0$ and $O \to XO0$ leave the non-terminal unchanged.
- A symbol 1 placed on the right flips it, which is why $E \to XO1$ and $O \to XE1$ switch between the two non-terminals.
- The base case $E \to \varepsilon$ is correct because the empty second half contains zero 1s, and O has no ε-rule for the same reason.

Formally, both inclusions follow by induction on n. Each rule preserves the claimed description, as the previous paragraph shows. For the other direction, let xy have $|x| = |y| = n$.

- If $n = 0$, then $\#(1, y) = 0$ is even and $E \to \varepsilon$ applies.
- If $n \geq 1$, write $x = ax'$ and $y = y'b$ with $a, b \in \Sigma$. Then $|x'| = |y'| = n - 1$, and $\#(1, y) = \#(1, y') + 1$ when $b = 1$, while $\#(1, y) = \#(1, y')$ when $b = 0$. So if $\#(1, y)$ is even, then $x'y'$ is generated by E when $b = 0$ and by O when $b = 1$, and the rules $E \to XE0$ and $E \to XO1$ produce xy. The odd case is symmetric.

Sources and collaborators: None.

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Part (c)

$$\{0^a1^b0^c1^d \mid a, b, c, d \geq 1 \text{ and if } a = d \text{ then } b = c\}$$

$S \to OMI \mid G \mid H$	the language
$G \to 0G1 \mid 0OIO1$	$0^a1^b0^c1^d$ with $a > d$
$H \to 0H1 \mid 0IOI1$	$0^a1^b0^c1^d$ with $a < d$
$M \to 10 \mid 1M0$	1^n0^n with $n \geq 1$
$O \to 0 \mid 0O$	0^n with $n \geq 1$
$I \to 1 \mid 1I$	1^n with $n \geq 1$

The six non-terminals generate the following languages.

- O generates every non-empty string of 0s, that is, every 0^n with $n \geq 1$.
- I generates every non-empty string of 1s, that is, every 1^n with $n \geq 1$.
- M generates every string 1^n0^n with $n \geq 1$.
- G generates every string $0^a1^b0^c1^d$ with $a, b, c, d \geq 1$ and $a > d$.
- H generates every string $0^a1^b0^c1^d$ with $a, b, c, d \geq 1$ and $a < d$.
- S generates every string $0^a1^b0^c1^d$ with $a, b, c, d \geq 1$ such that $b = c$ whenever $a = d$, which is the language asked for.

Why this is correct. The condition "if $a = d$ then $b = c$" is the same as "$a \neq d$ or $b = c$", so the language is the union of three sets: the strings with $b = c$, the strings with $a > d$, and the strings with $a < d$. The three alternatives for S generate exactly those three sets.

- OMI generates $0^a1^n0^n1^d$ for all $a, n, d \geq 1$, which is exactly the set of strings of the required shape with $b = c$.
- A derivation from G applies $G \to 0G1$ some $k \geq 0$ times and then $G \to 0OIO1$, producing

  $$0^k(0\,0^p1^q0^r1)1^k = 0^{k+1+p}1^q0^r1^{k+1}$$

  for some $p, q, r \geq 1$. So $a = k + 1 + p$, $b = q$, $c = r$, $d = k + 1$, and $a - d = p \geq 1$. Conversely, any string $0^a1^b0^c1^d$ with $b, c \geq 1$ and $a > d \geq 1$ arises this way, by taking $k = d - 1$, $p = a - d$, $q = b$, and $r = c$.
- A derivation from H likewise produces

  $$0^k(0\,1^p0^q1^r1)1^k = 0^{k+1}1^p0^q1^{r+1+k}$$

  for some $p, q, r \geq 1$. So $a = k + 1$, $b = p$, $c = q$, $d = k + 1 + r$, and $d - a = r \geq 1$. Conversely, any string with $b, c \geq 1$ and $d > a \geq 1$ arises by taking $k = a - 1$, $p = b$, $q = c$, and $r = d - a$.

Every string generated by S therefore has the shape $0^a1^b0^c1^d$ with all four exponents at least 1, and satisfies $a \neq d$ or $b = c$, so it is in the language. Conversely, a string of the language with $a = d$ must have $b = c$ and is generated by OMI, and a string with $a \neq d$ is generated by G or by H. The grammar is ambiguous, since strings with $b = c$ and $a \neq d$ have derivations of both kinds, but that is permitted here.

Sources and collaborators: None.

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